Sahaj Bhadja

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Summary

Software engineer with 2+ years building cross-platform systems, simulation pipelines, real-time applications, and production services. Experienced in C++, Python, spatial modeling, platform integration, automated testing, performance debugging, CI/CD, and reliable delivery across Windows, Linux, and macOS.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python platform services and interactive dashboards for 500+ users using Azure SQL and Blob Storage, tuning indexed queries and backend access paths to reduce latency from 1.2 seconds to 350 milliseconds.
- Implemented concurrent workflows for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving runtime correctness, observability, maintainability, and production issue investigation.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, integrating secure uploads, authentication, job-state updates, and grading workflows while investigating production issues for 150+ users.
- Created multithreaded processing for 1,000+ files with validation, AWS RDS persistence, progress updates, and failure handling, reducing turnaround 60% while maintaining consistent behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible PyTorch simulation pipelines processing 100,000+ trajectory logs, applying automated evaluation, performance instrumentation, and systematic debugging to improve long-horizon planning accuracy 18% across diverse controlled scenarios.
- Developed a spatial-temporal encoder integrating map geometry and agent context through multi-head attention, improving closed-loop stability 15% while validating correctness and performance across repeated controlled simulation environments.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed cross-platform Python services and REST integrations using AWS Glue, S3, SQL, filtering, and pagination, improving response times 30% and data-processing efficiency 68% across diverse production environments.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating platform issue diagnosis, release validation, and reliable production deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight services with reusable interfaces, JWT authentication, and WebSocket updates, synchronizing 2 million+ Azure SQL records through consistent platform APIs.
- Automated Docker builds, functional tests, staged deployments, and runtime monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, platform consistency, and operational visibility.

Skills

Languages & Systems: C++, Python, Java, JavaScript, TypeScript, SQL, object-oriented programming, data structures, algorithms, concurrency, multithreading
Platform & Integration: Windows, Linux, macOS, React, Node.js, Flask, FastAPI, REST APIs, WebSockets, platform APIs, distributed workflows, spatial modeling
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, AWS RDS, AWS, Azure, S3, Blob Storage, data modeling, query optimization
Quality & Delivery: Git, Docker, Kubernetes, Jenkins, CI/CD, PyTest, Mockito, automated testing, unit testing, integration testing, simulation, benchmarking, debugging, instrumentation, monitoring, code review, Agile

Projects

CPU-Scheduling Simulator | Live | *C++, Simulation, Concurrency, Operating Systems, React*
- Implemented six scheduling algorithms in C++ with deterministic simulations and performance visualization, evaluating waiting and turnaround behavior across 1,000+ process configurations while reducing systematic debugging time 20%.

PiXMod | Live | *Next.js, TypeScript, MongoDB, Stripe, CLIP, SAM*
- Built an interactive Next.js and TypeScript image-editing application using CLIP and SAM, implementing reusable interface components, secure authentication, MongoDB persistence, and responsive computer-vision workflows across multiple devices.

FinNexus | Live | *TypeScript, React, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built real-time financial workflows supporting 50,000+ users and one million transactions, integrating REST APIs, WebSockets, MongoDB persistence, validation, monitoring, and automated Kubernetes deployment through reliable CI/CD pipelines.